Do Now · retrieval

Floating point → denary

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Represent 6.5 in normalised floating-point form using an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent.

Q2

Convert $6D_{16}$ to eight-bit binary and denary.

Q3

Convert 11110110 from 8-bit two’s-complement binary to denary. Show signed place values.

Q4

Apply AND 00001111 to 10110110 . Give the result in binary and hexadecimal.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Split the fields — Separate the bit pattern into the mantissa and the exponent at the stated field boundary.

Core idea 2

Evaluate the mantissa — Add the place values for the bits that are 1. — CODE:
Mantissa places:
−1 | ½ ¼ ⅛ 1⁄16 1⁄32 1⁄64 1⁄128

Core idea 3

Convert the exponent to denary — Read the exponent as a signed two’s-complement integer . — CODE:
4-bit exponent places:
−8 4 2 1

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Apply the scale — CODE:
Value = mantissa × 2^{E} — A positive exponent multiplies by a power of two. A negative exponent divides by a power of two.

Core idea 5

Check the representation — For a non-zero normalised mantissa: — 01 starts a positive normalised mantissa.

Vocabulary · use the correct technical terms

Mantissa — Which part of a floating-point number stores the signed fractional value?

Exponent — Which part controls the scale of a floating-point value?

Normalised — What word describes a non-zero mantissa arranged to begin 01 if positive or 10 if negative?

Precision — More mantissa bits improve the _____ of a floating-point representation.

Independent skill check · apply the lesson

Decode 01110000 0010 to denary (8-bit mantissa, 4-bit exponent). Is 00011010 0101 normalised? Justify.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Convert this floating-point representation to denary

Follow the worked method

1. Evaluate the mantissa — CODE:
01011000
= ½ + ⅛ + 1⁄16
= 0.6875 — Read the mantissa as a signed binary fraction.

2. Convert the exponent to denary — CODE:
0011 = 3 — The exponent is a signed two’s-complement integer.

3. Apply the scale — CODE:
0.6875×2^3
$= 0.6875 \times 8$
$= 5.5$

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · solo guided practice

Work ALONE. Try first; read the clues one at a time if you get stuck.

Complete the steps below. Check your answer only on page 8.

YOUR TASK

Build the answer

CLUES · uncover one at a time

Clue 1: For an 8-bit two's-complement mantissa, weights begin -1, 1/2, 1/4, 1/8.

Clue 2: Decode the four-bit exponent as a signed integer.

Clue 3: Multiply the signed mantissa by 2 to the power of the exponent.

BUILD YOUR ANSWER · in BLACK PEN

Step 1: Decode the mantissa.

Step 2: Decode the exponent.

Step 3: Scale and calculate the final denary value.

CHECK BEFORE TURNING THE PAGE

Have you shown every step? The worked self-check is at the END on page 8.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Convert 0100 1110 to denary

The mantissa is four bits and the exponent is four bits. Both use two’s complement. Show your working.

Paraphrased OCR June 2023 Q3(d); original values and marks retained. Source .

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Precision and range
What improves if more bits are allocated to (a) the mantissa and (b) the exponent?
Then explain the trade-off if the total number of bits is fixed.
Think about what mantissa bits control and what exponent bits control.
The first two prompts paraphrase OCR June 2023 Q3(e). The trade-off prompt is an original classroom extension. Source .

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A negative exponent means the floating-point value must be negative.

Use key vocabulary from memory

1. Explain Mantissa in your own words.

2. Explain Exponent in your own words.

3. Explain Normalised in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

END SELF-CHECK · only after you have attempted the work

PAGE 5 · Guided We Do solution:

10110000 = -0.625; 0010 = +2; -0.625 x 2^2 = -2.5.

PAGE 3 · Independent skill check:

01110000 = +0.875, exponent 0010 = +2, so 3.5. 00011010 is not normalised: its first two mantissa bits are 00.

Correct any errors in GREEN PEN. Write one improvement.

YOU DO ANSWERS: rhsccs.org/lessons/h446/lesson-005/answers.html